

Solid Propellant

Solid-propellant rockets are the oldest type of rocket. Relatively simple in construction as shown in Fig. 1. Solid rocket engines are used on air-to-air and air-to-ground missiles and on model rockets and as **boosters** for satellite launchers. A simple solid rocket motor consists of a casing, propellant charge (identified as grain), igniter, and nozzle. This grain contains both solid fuel and solid oxidizer components combined within a cylindrical combustion chamber or case. The propellant is casted into the rocket shell having a central cavity of different shapes including star shaped that serves as combustion chamber

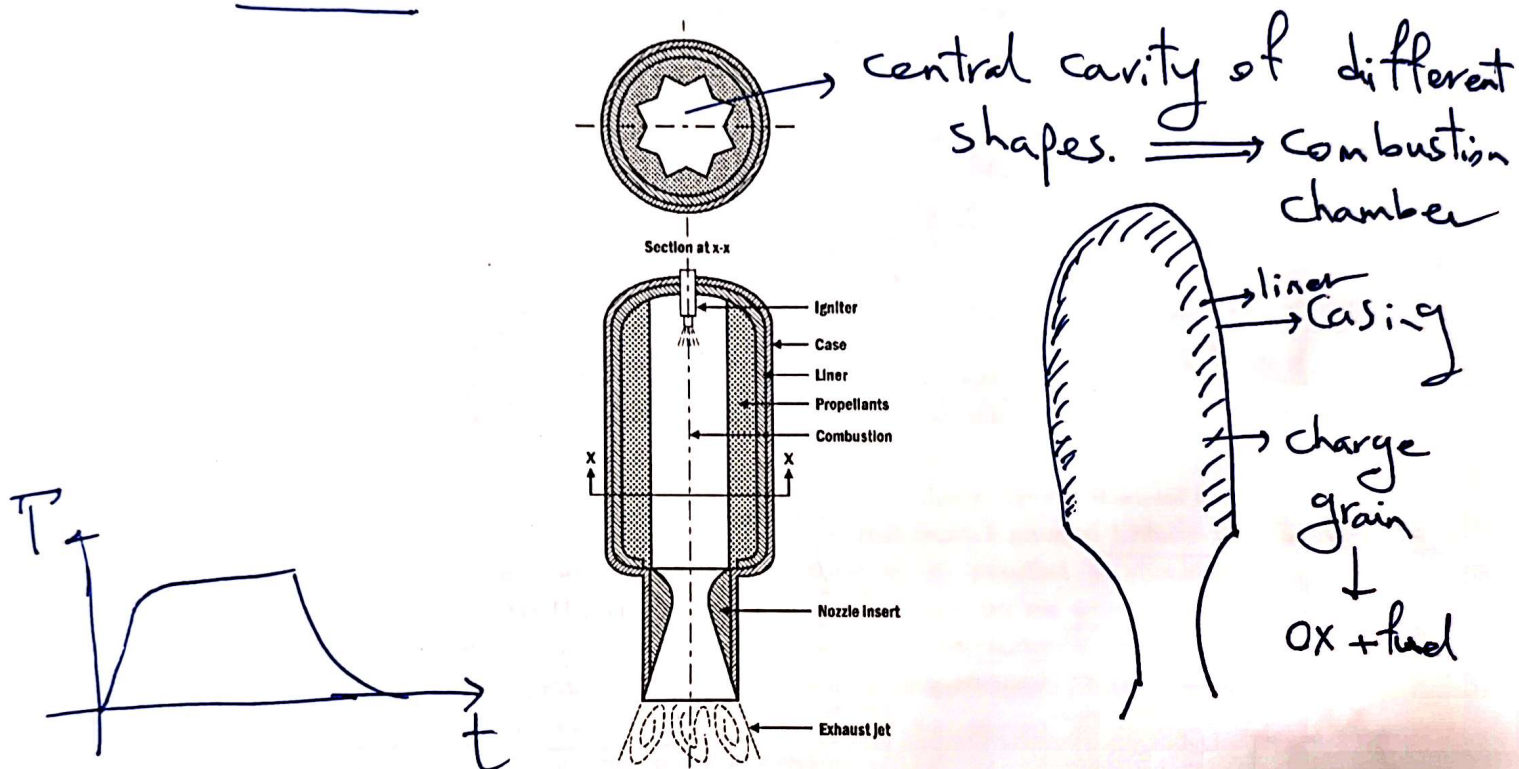


Fig. 1: Solid propellant rocket

Thrust-burning time profile depends on this cavity. A liner provided between the case and the propellant protects the case from high temperatures developing inside the propellant layers.

The liner is a protective layer that is applied between the propellant grain (solid fuel) and the insulation or motor case.

Liners are typically made from polymeric materials, such as epoxy-based resins or polyurethane, which are chosen for their thermal resistance, adhesive properties, and chemical stability.

An electrical signal is sent to the igniter which creates hot gases that ignite the main propellant grain. Once the flame front is established, combustion is self sustaining. The rate of burning is proportional to the exposed surface area. As the propellant burns, a "flame front" is produced which moves into the propellant until all the propellant is burned. The hot exhaust gases produced in combustion are used to propel the rocket.

$i \propto \text{Area}$

220 kft



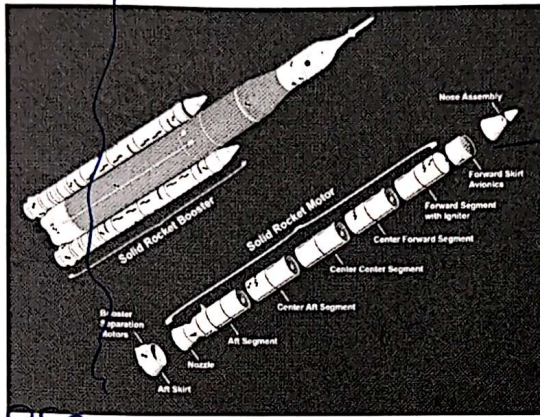
Solid propellant must have the following features:

1. High chemical energy to generate maximum thrust and specific impulse
2. High density to enable packing a large quantity of propellant in a small volume
3. Low molecular weight thus the exhaust gases achieve high acceleration
4. Easy to ignite
5. Burns steadily at predictive rate
6. Easy to fabricate
7. Smoke-free and nontoxic

$$u_e \propto \sqrt{\frac{Q}{M}}$$

$h_f \uparrow$
 $g \uparrow$
 $M_w \downarrow$

150, kft
burning ends



Solid rocket

12 ft
150 ft

Pacific Ocean

Fig. 2: Space Shuttle Solid Rocket Boosters

Space Shuttle Solid Rocket Boosters (SRBs) are the largest solid-propellant motors ever flown and first designed for reuse. Each measures 12.17-ft diameter and 149.16-ft length. It generates 3,300,000 lb sea-level thrust. Its weight with propellant is 1,300,000 lb. Its propellant is ammonium perchlorate/aluminum and has 11 point star shape in forward motor segments and double truncated cone in aft segments. Burn time is 75 s and its expansion ratio is 11.3. After burnout at approximately 150,000 ft, the spent cases separate from the vehicle and arcing up to approximately 220,000 ft before parachuting to the ocean for recovery and reuse.

3.3M lb
AP/AL

Table 1 presents advantages and disadvantages of solid rockets.

Table 1: Advantages and disadvantages of solid rockets

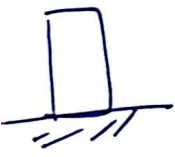
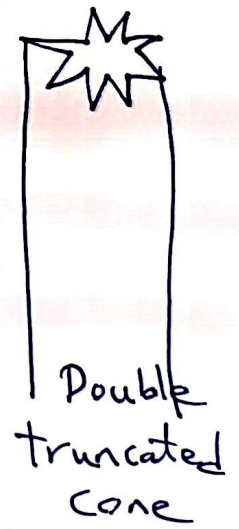
Advantages	Disadvantages
Compact and smaller in size compared with liquid rocket	Once ignited cannot be shut off Used only once
Do not require propellant feeding system or turbopumps as liquid rocket	
High thrust-to-weight ratio	
Require little maintenance	
Safe and reliable and thus can sit for years before firing	

Composition of a Solid Propellant

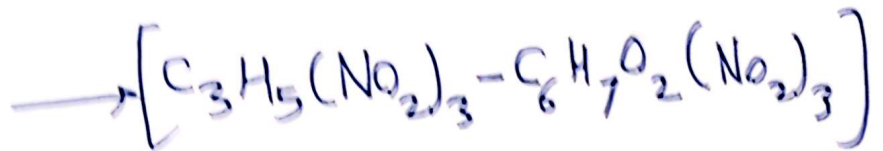
There are basically three types of propellant grains, namely:

1. Homogeneous or colloidal propellants

Fuel and oxidizer are contained in the same molecule which decomposes during combustion. Typical examples are nitroglycerine (NG) and nitrocellulose (NC). Most propellants are double based (few single-based and triple-based propellants are found). Double-based propellants are combinations of



Homogeneous
NG-NC



nitroglycerine (NG) and nitrocellulose (NC) $[C_3H_5(NO_2)_3 - C_6H_7O_2(NO_2)_3]$ with small quantity of additives. It has nontoxic and smokeless exhausts.

1. Nitroglycerine (NG):

- **Role:** Nitroglycerine is a powerful liquid explosive and is used as a plasticizer and energy contributor in solid rocket propellants.
- **Function in Solid Propellants:** In solid propellants, NG is mixed with nitrocellulose to form a homogeneous material. It contributes to the energy density of the propellant, providing a large amount of energy during combustion. NG also helps to make the propellant more flexible and less brittle.
- **Combustion:** When NG burns, it produces a high amount of heat and gas, which leads to the rapid expulsion of gas through the rocket nozzle, generating thrust.

2. Nitrocellulose (NC):

- **Role:** Nitrocellulose is a flammable polymer that serves as both a binder and a fuel in the propellant.
- **Function in Solid Propellants:** NC is used to hold the propellant mixture together and provides structural integrity. It also acts as a combustible material that contributes to the overall energy release during the rocket's operation. NC typically forms the bulk of the propellant and stabilizes the propellant mass.
- **Combustion:** Upon ignition, NC decomposes into gases and heat, which aids in the continuous burning of the propellant. It also plays a crucial role in controlling the burn rate of the propellant.

2. Composite or heterogeneous propellants

These are mixtures of oxidizing crystals and a powdered fuel (usually aluminum) held together in a binder (synthetic rubber or plastic). Sometimes light metal powders are added to increase the energy of the combustion process and fuel density. They are more stable than homogeneous and preferred in long-term stored rockets. However, the addition of light metals makes their exhaust toxic and smoky.

3. Composite modified double-based propellants

They are a heterogeneous combination of the double-based homogeneous (colloidal) propellants and composite propellants.

Ox-crystal + powdered fuel (AL)
rubber